



Computer Networks 03 | Types of Communication | Computer Science & IT | GATE Crash Course

Generated on October 23, 2024

Summary

AI Notes

AI Slides

Text Notes

Screenshots

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📷 29

class A $\rightarrow 2^{24}$ IP Addresses in one n/w
class-B $\rightarrow 2^{16}$ IP Addresses in one n/w
class-C $\rightarrow 2^8$ IP " " "

I: organization X Need $\rightarrow 2^{20}$ IP Addresses

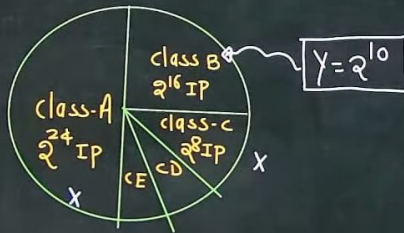
$X = 2^{20}$

class-A 2^{24} IP
class-B 2^{16} IP
class-C 2^8 IP

IP Address wasted
 $= 2^{24} - 2^{20}$
 $= 2^4 \times 2^{20} - 2^{20}$
 $= 16 \times 2^{20} - 2^{20}$
 $= 15 \times 2^{20}$
 $= 15M$
 $= 15,728$

▶ 19:39

II organization Y Need = 2^{10} IP Addresses

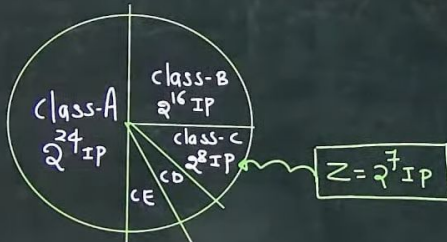


IP Address wasted

$$\begin{aligned}
 &= 2^{16} - 2^{10} \\
 &= 2^6 * 2^{10} - 2^{10} \\
 &= 64 * 2^{10} - 2^{10} \\
 &= 63 * 2^{10} \\
 &= 63K \\
 &= 64,512
 \end{aligned}$$

▶ 22:54

III organization Z need = 2^7 IP Addresses



IP Address wasted

$$\begin{aligned}
 &= 2^8 - 2^7 \\
 &= 2 * 2^7 - 2^7 \\
 &= 2^7 = 128
 \end{aligned}$$

▶ 25:41

Q.13

What is the possible number of networks and addresses in each network under class B addresses in IPv4 addressing format .



A

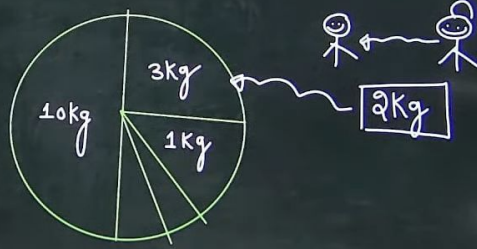
$2^{16}, 2^{16}$

B

C

D

GATE WALLAH



▶ 28:14

Q.13

What is the possible number of networks and addresses in each network under class B addresses in IPv4 addressing format .



A

$2^{16}, 2^{16}$

B

$2^{16}, 2^{16} - 2$

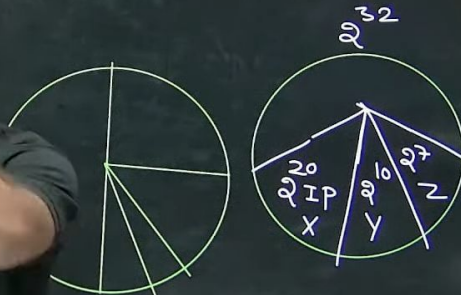
C

$2^{14}, 2^{14}$

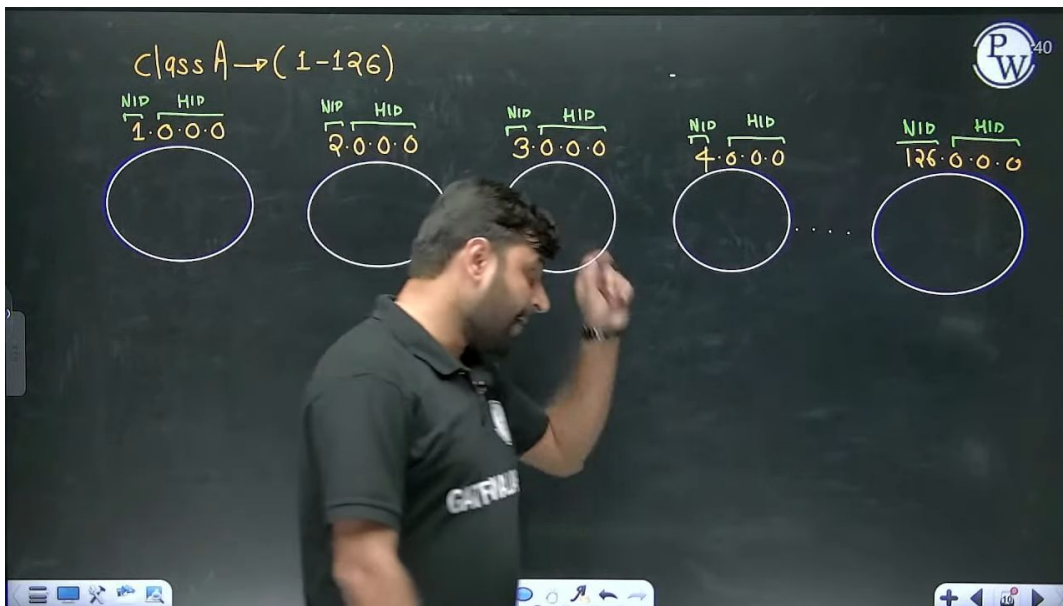
D

$2^{14}, 2^{14} - 2$

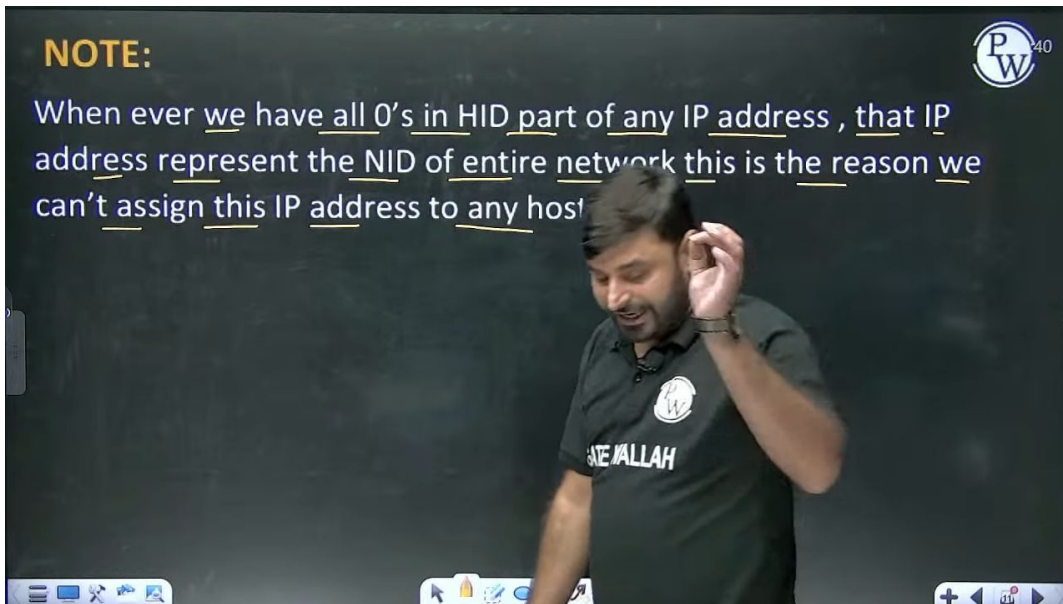
GATE WALLAH



▶ 31:52



▶ 37:50



▶ 38:20

① IP Add = $\frac{10}{\text{NID}} \cdot \frac{32}{\text{HID}} \cdot 92 \cdot 108 \rightarrow \text{class A}$
NID = $10 \cdot 0 \cdot 0 \cdot 0$

② IP Add = $\frac{157}{\text{NID}} \cdot \frac{163}{\text{HID}} \cdot 92 \cdot 97$
NID = $157 \cdot 163 \cdot 0 \cdot 0$

10.0.0.0

157.163.0.0

class-B

PW⁴⁶

▶ 43:45

③ IP Add = $\frac{200}{\text{NID}} \cdot \frac{139}{\text{HID}} \cdot 24 \cdot 67 \rightarrow \text{class-c}$
NID = $200 \cdot 139 \cdot 24 \cdot 0$

200.139.24

PW⁴⁷

▶ 45:17

TYPES OF COMMUNICATION



- (i) Unicast communication (1:1)
- (ii) Broadcast communication (1:All)
- (iii) Multicast Communication (1:Many)

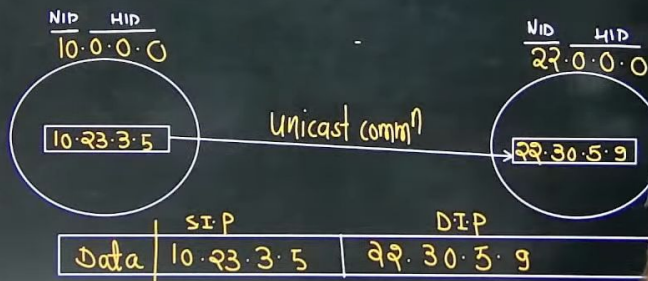
▶ 46:04

Unicast Communication:

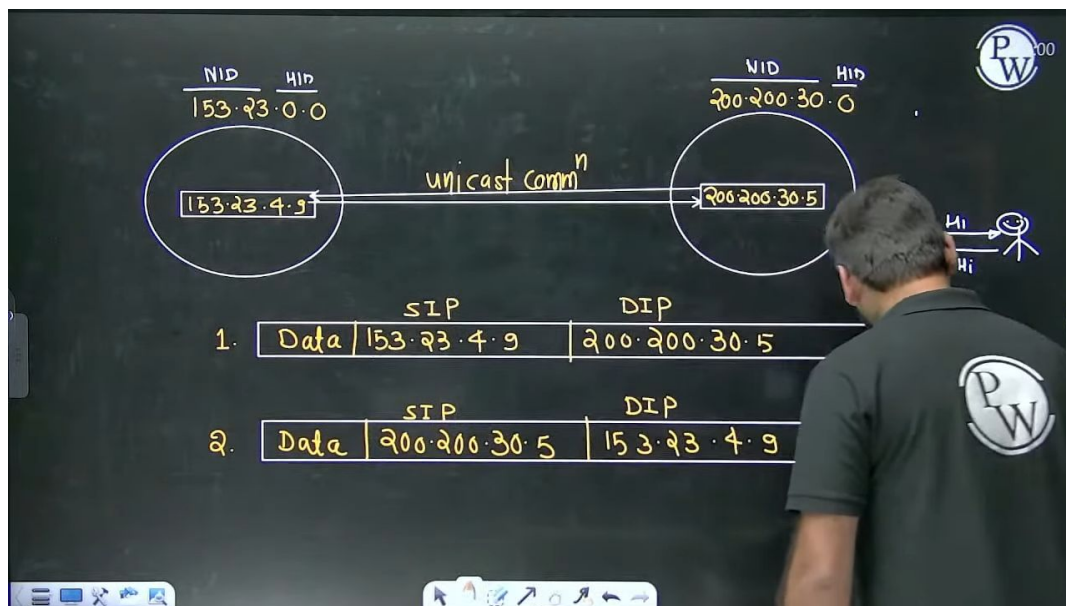


Transmitting the data from one computer to another computer is called as unicast communication.

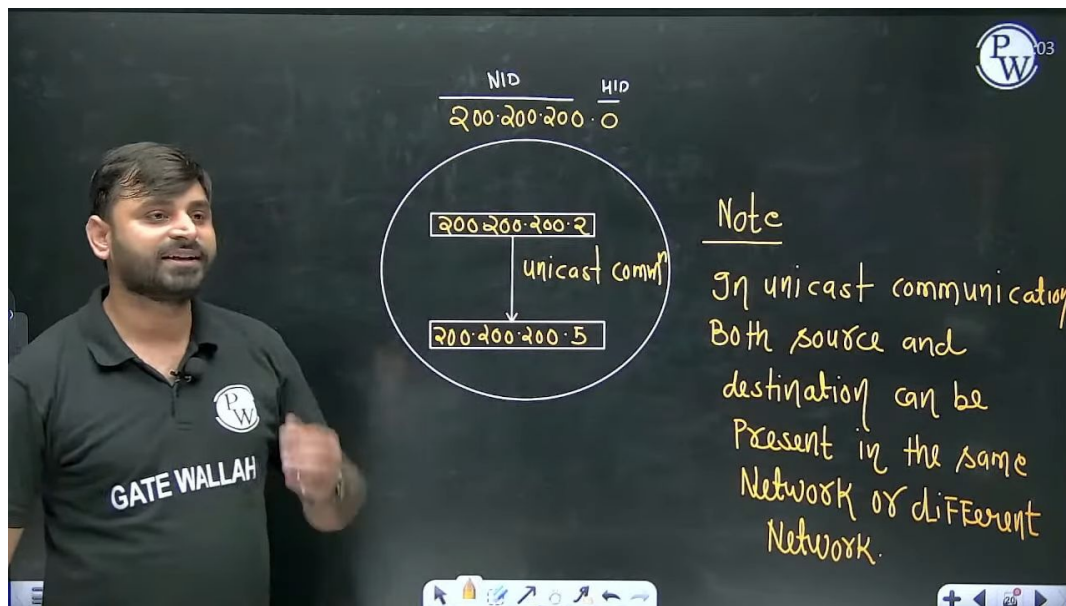
It is one to one transmission.



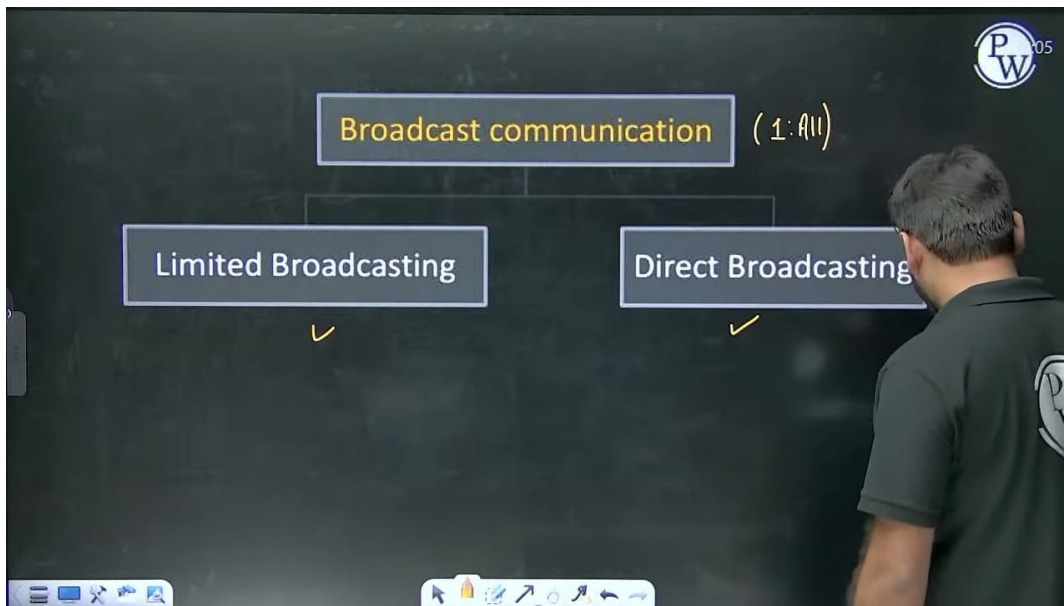
▶ 51:28



▶ 58:08



▶ 1:01:20



▶ 1:03:00

PW¹¹

Limited Broadcasting:

Transmitting data from one computer to all other computer in the same network is called as Limited Broadcasting.

Mid: 15.0.0.0 H1D: 15.0.0.0

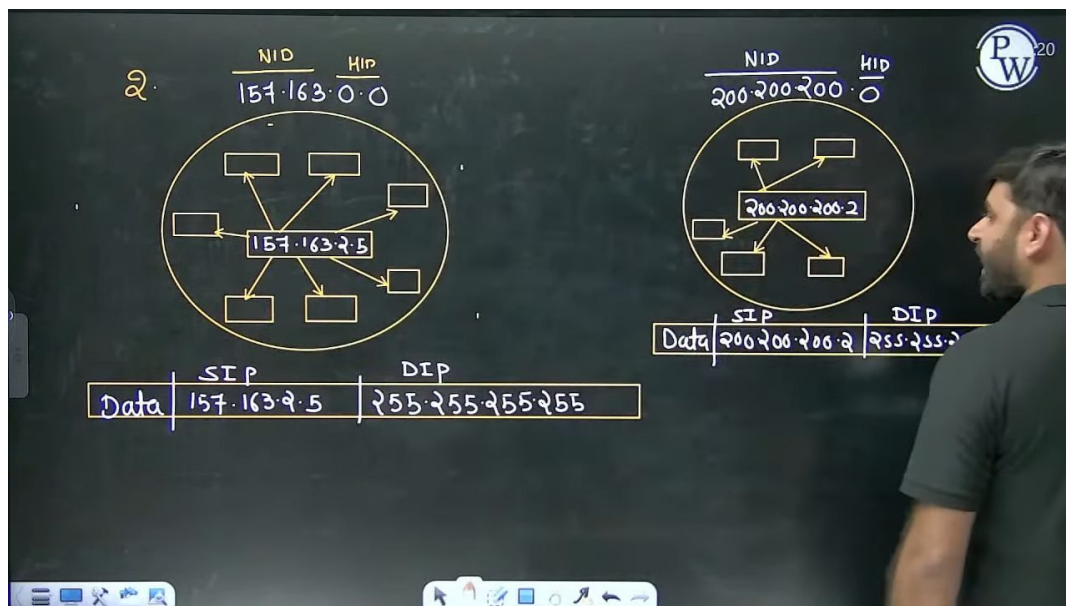
①

	S.I.P	D.I.P
Data	15.23.9.5	255.255.255.255

②

	S.I.P	D.I.P
Data	255.255.255.255	15.23.9.5

▶ 1:08:51

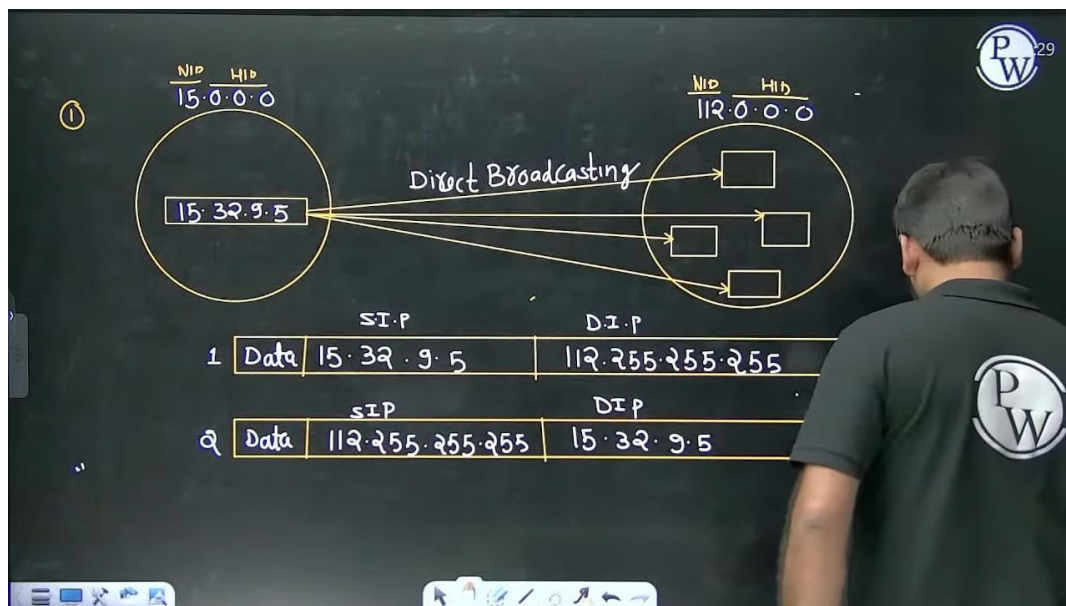


▶ 1:17:40

Direct Broadcasting:

Transmitting data from one computer to all other computer in the different N/w is called as Direct Broadcasting.

▶ 1:17:41



▶ 1:26:58

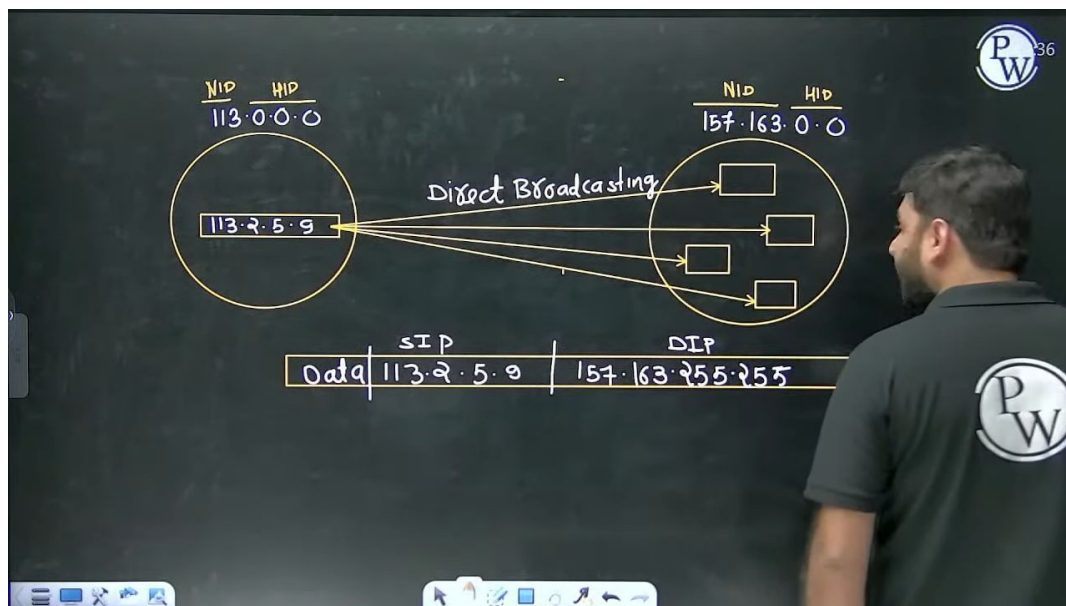
NOTE:

When ever we have all 1's in HID part of any IP address , that IP address represent the Direct broadcast address so this is the reason we can't assign this IP address to any host.

Note: ① Direct Broadcast Address can't be used as a SIP Address

② Direct Broadcast Address will Always be used as DIP Address.

▶ 1:28:55



▶ 1:33:42

Table illustrating IP Address Validity:

NID	HID
Valid	0's → NID of entire Network
Valid	1's → Direct Broadcast Address
1's	1's → Limited Broadcast Address

▶ 1:36:37

	IP Address	Network-Id	Direct Broadcast Address	Limited Broadcast Address
Class A	19.35.21.31 NID: 19.35.21 HID: 31	19.0.0.0	19.255.255.255	255.255.255.255
Class B	119.31.34.2 NID: 119.31 HID: 34.2	119.0.0.0	119.255.255.255	255.255.255.255
Class B	150.0.94.31 NID: 150.0 HID: 94.31	150.0.0.0	150.0.255.255	"
Class B	190.34.17.31 NID: 190.34 HID: 17.31	190.34.0.0	190.34.255.255	"
Class C	200.200.34.92 NID: 200.200.34 HID: 92	200.200.34.0	200.200.34.255	"
Class C	217.39.47.9 NID: 217.39.47 HID: 9	217.39.47.0	217.39.47.255	"
Class D	226.9.7.97 NID: 226.9.7 HID: 97	X	X	X
Class E	243.2.3.5 NID: 243.2.3 HID: 5	X	X	X

▶ 1:46:16

Network Masks:

A network mask helps you to know which portion of the address identifies the network-id and which portion of the address identifies the host-id. Class A, B, and C networks have default masks, also known as natural masks, as shown here:

Class A: 255.0.0.0

Class B: 255.255.0.0

Class C: 255.255.255.0

▶ 1:48:11

Note: In the Network Mask No. of 1's indicate NID Part
and No. of 0's indicate HID Part

Class A: 11111111. 00000000. 00000000. 00000000
255.0.0.0

Class B: 11111111. 11111111. 00000000. 00000000
255.255.0.0

Class C: 11111111. 11111111. 11111111. 00000000
255.255.255.0

▶ 1:53:10

IP Address = 200.200.200.96

Network Mask = 255.255.255.0

IP Address: 11001000. 11001000. 11001000. 01100000

Mask = 11111111. 11111111. 11111111. 00000000
NID HID

NID = 200.200.200

HID = 96

▶ 1:56:46

IP Address : 11001000 · 11001000 · 11001000 · 01100000

ANDing AND

Network Mask 11111111 · 11111111 · 11111111 · 00000000

NID 11001000 · 11001000 · 11001000 · 00000000

NID = 200 · 200 · 200 · 0

IP Addr = $\frac{\text{NID}}{200 \cdot 200 \cdot 200 \cdot 96}$

NID = 200 · 200 · 200 · 0

shortcut

Any 8 bit Number
ANDing
255
Any Number

▶ 2:03:16

Q.3

What is the network ID (NID) of the IP address 230.100.123.70 ? (Assuming Classfull addressing scheme is followed.)

- A 230.100.123.0
- B 230.100.0.0
- C 230.0.0.0
- D None of these

GATE WALLAH

▶ 2:08:16

Q.5 ^{class-A}100.86.95.75, ^{class B}157.192.190.253, ^{class-C}200.1.56.97, ^{class A}10.34.87.95. which is common for all these IP Addresses.

- ☐ A Class of IP address
- ☐ B Limited broadcast address
- ☐ C Network address
- ☐ D Direct broadcast address



▶ 2:13:28

Q.5 ^{class-A}100.86.95.75, ^{class B}157.192.190.253, ^{class-C}200.1.56.97, ^{class A}10.34.87.95. which is common for all these IP Addresses.

- ☒ A Class of IP address
- ☐ B Limited broadcast address
- ☐ C Network address
- ☐ D Direct broadcast address



▶ 2:15:02